

TASK 4

Implement the encryption of block chunk of 128 bits size using AES algorithm in C/JAVA

Aim: To implement AES Algorithm logic.

Program:

```
import java.security.*;
import java.io.*;

import javax.crypto.*;
import java.util.*;

public class Task4 {

    static Scanner s = new
    Scanner(System.in); private static Cipher
    cipher;

    private static byte[] encrypt(String data) throws Exception{
        byte[] encrypted= cipher.doFinal(data.getBytes());

        System.out.println(encrypted);
        return encrypted;
    }

    private static byte[] decrypt(byte[] data) throws Exception{
        byte[] decrypted= cipher.doFinal(data);

        System.out.println(new
        String(decrypted)); return decrypted;
    }

    public static void main(String[] args)
    { try {

        String data = s.nextLine();

        KeyGenerator keygen = KeyGenerator.getInstance("AES");
        SecretKey key = keygen.generateKey();

        cipher=Cipher.getInstance("AES");
        cipher.init(Cipher.ENCRYPT_MODE
        ,key); byte[] encrypted =
        encrypt(data);

        cipher.init(Cipher.DECRYPT_MODE,key);
        decrypt(encrypted);

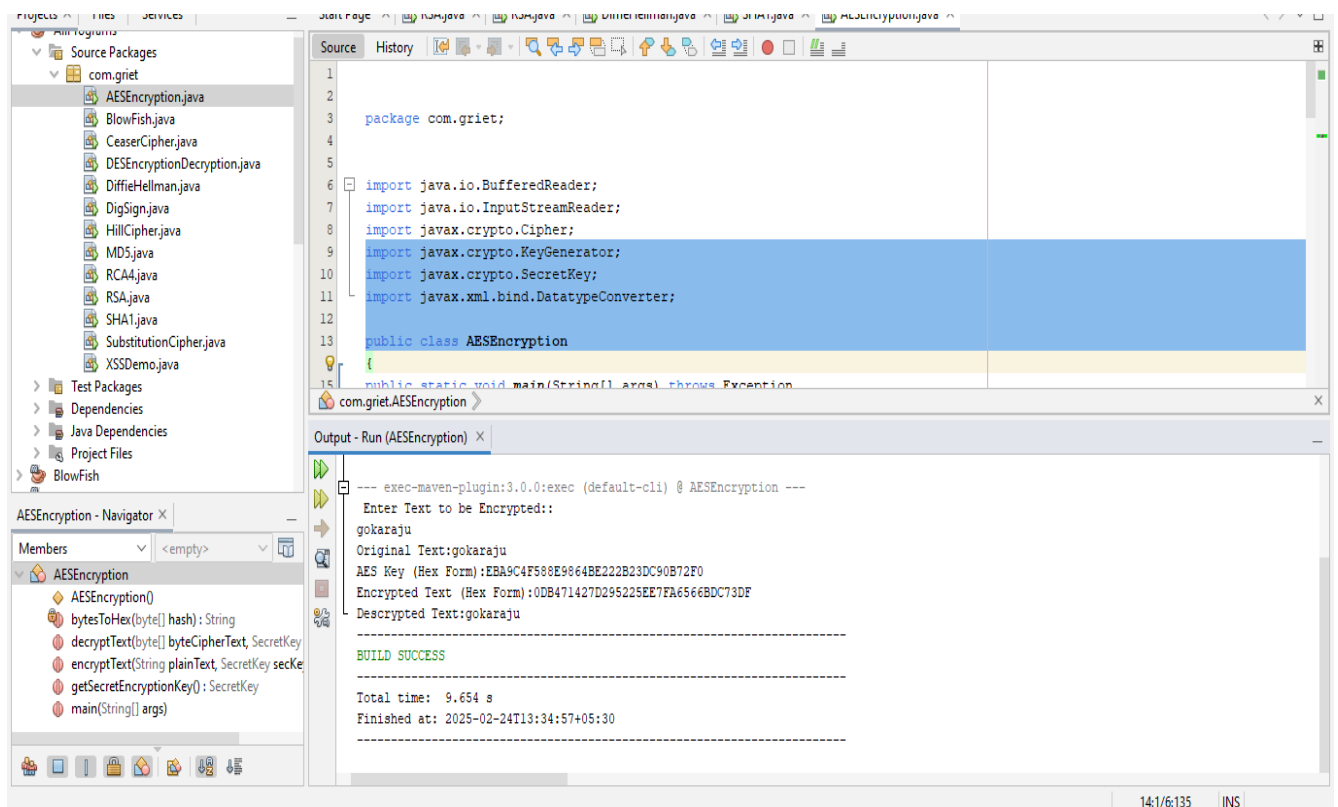
    } catch (Exception e)
```

```

    {
        e.printStackTrace();
    }
}
}
}

```

OUTPUT:



VIVA QUESTIONS

1. What is AES, and why was it chosen as the standard encryption algorithm?
2. How does AES perform encryption and decryption, and what are its key components?
3. What are the different key sizes used in AES, and how do they affect security?
4. How does AES resist cryptographic attacks like brute force and differential cryptanalysis?